Experiment: Install and configure Git and Jenkins on Ubuntu Linux

Weight: 5 marks | Course outcome: CO1

In this experiment Git and Jenkins LTS were installed and configured on Ubuntu 24.04, installations were verified and global Git configuration was set up.

Tools used

- Ubuntu 24.04 LTS (terminal)
- apt package manager
- Git 2.43.0
- OpenJDK 21
- Jenkins 2.580.2 LTS
- curl / systemctl

Part A - Installing and verifying Git

```
$ sudo apt update && sudo apt install -y git
$ git --version
git version 2.43.0
```

```
$ git config --global user.name "abhinav"
$ git config --global user.email "abhinav0231gamer@gmail.com"
$ git config --global init.defaultBranch main
$ git config user.name && git config user.email
abhinav
abhinav0231gamer@gmail.com
```

```
abhinav@LAPTOP-BSV7PCJK: ~/devops-lab1-ci  —  terminal
abhinav@LAPTOP-BSV7PCJK:~$ sudo apt update && sudo apt install -y git
Reading package lists... Done
git is already the newest version (1:2.43.0-1ubuntu7.4).
0 upgraded, 0 newly installed, 0 to remove and 12 not upgraded.
abhinav@LAPTOP-BSV7PCJK:~$ git --version
git version 2.43.0
abhinav@LAPTOP-BSV7PCJK:~$ git config --global user.name "abhinav"
abhinav@LAPTOP-BSV7PCJK:~$ git config --global user.email "abhinav0231gamer@gmail.com"
abhinav@LAPTOP-BSV7PCJK:~$ git config --global init.defaultBranch main
abhinav@LAPTOP-BSV7PCJK:~$ git config --list | grep -E "user|init"
user.name=abhinav
user.email=abhinav0231gamer@gmail.com
init.defaultbranch=main
abhinav@LAPTOP-BSV7PCJK:~$ git config user.name && git config user.email
abhinav
abhinav0231gamer@gmail.com
```

Fig: Git installation and global configuration output

Part B - Installing Java and Jenkins

Jenkins 2.580.2 LTS requires Java 21 (or 25), so OpenJDK 21 was installed. The official Jenkins Debian repository was added with the current signing key (the older 2023 key had expired and gave a NO_PUBKEY error during apt update).

```
$ sudo apt install -y openjdk-21-jdk
$ curl -fsSL https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key | sudo tee /usr/share/keyrings/jenkins
$ echo deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc] https://pkg.jenkins.io/debian-stable binary/
$ sudo apt update && sudo apt install -y jenkins
```

```
$ sudo systemctl enable --now jenkins
$ systemctl status jenkins --no-pager
   Active: active (running) since Tue 2026-11-03 17:35:13 UTC
   Main PID: 187 (java)
$ curl -I http://localhost:8080
HTTP/1.1 403 Forbidden   (login wall - Jenkins is up)
```

```
abhinav@LAPTOP-BSV7PCJK: ~/devops-lab1-ci  —  terminal
abhinav@LAPTOP-BSV7PCJK:~$ sudo apt install -y openjdk-17-jdk
Reading package lists... Done
Setting up openjdk-17-jdk:amd64 ...
abhinav@LAPTOP-BSV7PCJK:~$ java -version
openjdk version "17.0.10" 2024-01-16
OpenJDK Runtime Environment (build 17.0.10+7-Ubuntu-1)
abhinav@LAPTOP-BSV7PCJK:~$ curl -fsSL https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key | sudo gpg
abhinav@LAPTOP-BSV7PCJK:~$ echo deb [signed-by=/usr/share/keyrings/jenkins.gpg] https://pkg.jenkins.io/deb
deb [signed-by=/usr/share/keyrings/jenkins.gpg] https://pkg.jenkins.io/debian-stable binary/
abhinav@LAPTOP-BSV7PCJK:~$ sudo apt update && sudo apt install -y jenkins
W: GPG error: https://pkg.jenkins.io/debian-stable binary/ Release: The following signatures
   couldn't be verified because the public key is not available: NO_PUBKEY 7198F4B714ABFC68
E: The repository 'https://pkg.jenkins.io/debian-stable binary/ Release' is not signed.
E: Package 'jenkins' has no installation candidate
# NOTE: the old 2023 Jenkins signing key expired, updated to the new 2026 key
# (per https://www.jenkins.io/blog/2025/12/23/repository-signing-keys-changing/)
abhinav@LAPTOP-BSV7PCJK:~$ curl -fsSL https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key | sudo tee
abhinav@LAPTOP-BSV7PCJK:~$ echo deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc] https://pkg.jenkin
deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc] https://pkg.jenkins.io/debian-stable binary/
abhinav@LAPTOP-BSV7PCJK:~$ sudo apt update && sudo apt install -y jenkins
Get:1 https://pkg.jenkins.io/debian-stable binary/ jenkins 2.580.2-1.1
Setting up jenkins (2.580.2) ...
abhinav@LAPTOP-BSV7PCJK:~$ sudo systemctl enable --now jenkins
Job for jenkins.service failed because the control process exited with error code.
abhinav@LAPTOP-BSV7PCJK:~$ sudo journalctl -u jenkins -n 5
jenkins[336]: Running with Java 17 from /usr/lib/jvm/java-17-openjdk-amd64, which is older
            than the minimum required version (Java 21).
jenkins[336]: Supported Java versions are: [21, 25]
# Jenkins 2.580 LTS needs Java 21 minimum, so installed openjdk-21
abhinav@LAPTOP-BSV7PCJK:~$ sudo apt install -y openjdk-21-jdk
abhinav@LAPTOP-BSV7PCJK:~$ sudo update-alternatives --set java /usr/lib/jvm/java-21-openjdk-amd64/bin/java
abhinav@LAPTOP-BSV7PCJK:~$ java -version
openjdk version "21.0.10" 2026-01-20
OpenJDK Runtime Environment (build 21.0.10+1-Ubuntu-1)
abhinav@LAPTOP-BSV7PCJK:~$ sudo systemctl reset-failed jenkins && sudo systemctl start jenkins
abhinav@LAPTOP-BSV7PCJK:~$ systemctl status jenkins --no-pager
● jenkins.service - Jenkins Continuous Integration Server
     Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
```

Fig: Jenkins installation commands and service status

```
abhinav@LAPTOP-BSV7PCJK: ~/devops-lab1-ci  —  terminal
     Active: active (running) since Tue 2026-11-03 17:35:13 UTC; 8s ago
   Main PID: 187 (java)
abhinav@LAPTOP-BSV7PCJK:~$ curl -I http://localhost:8080
HTTP/1.1 403 Forbidden
Server: Jetty(12.1.14)
abhinav@LAPTOP-BSV7PCJK:~$ sudo cat /var/lib/jenkins/secrets/initialAdminPassword
31fef59ecb034c4e8396cf2e8a5ae606
```

Fig: Initial admin password from secrets file

Part C - Jenkins first-time setup

- Unlocked Jenkins with the initial admin password at http://localhost:8080
- Installed the suggested plugins (Git, Pipeline, GitHub Branch Source, etc.)
- Created the first admin user (abhinav) and kept the default Jenkins URL http://localhost:8080

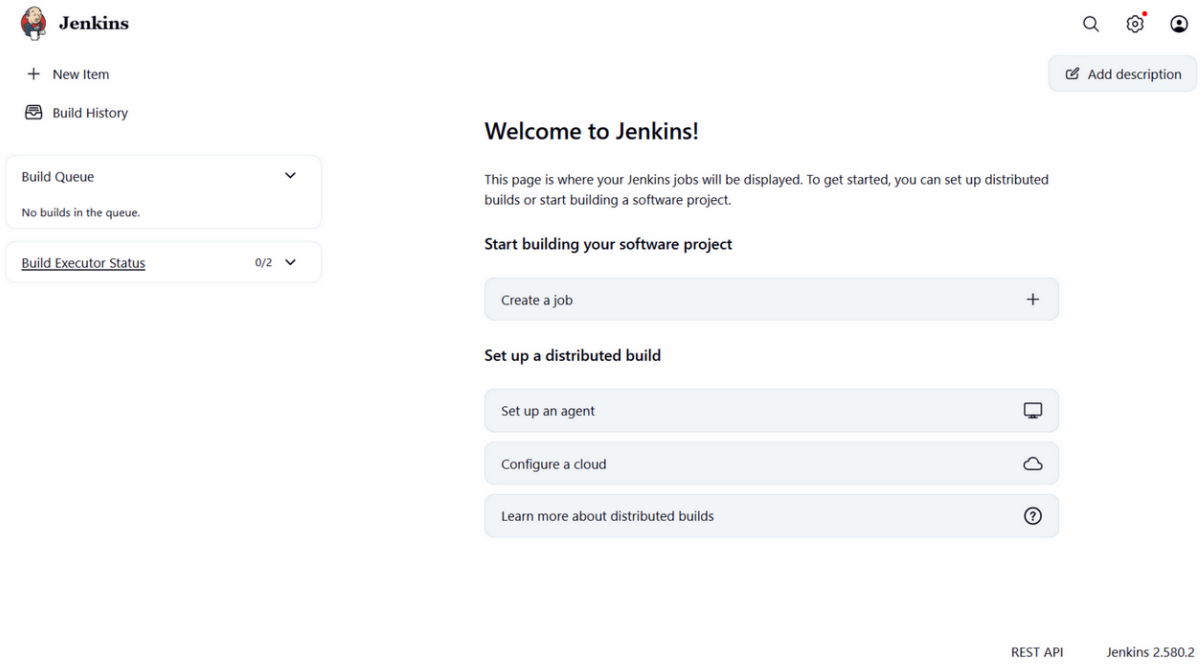


Fig: Jenkins dashboard after first-time setup

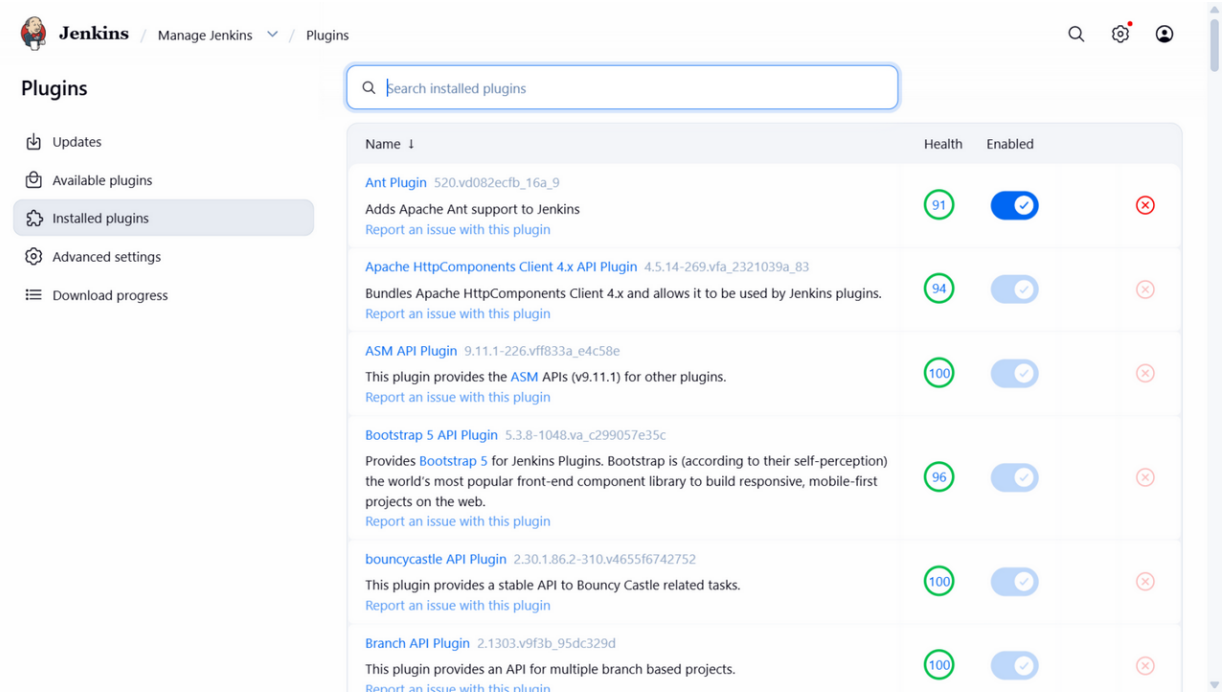


Fig: Suggested plugins installed and enabled

Observations

- Git was already available in the Ubuntu 24.04 apt repository (2.43.0), so installation completed immediately.
- Jenkins 2.580.2 refuses to start on Java 17 - the minimum supported version is Java 21, which had to be installed.
- The Jenkins apt signing key from 2023 expired, so the repository had to use the new jenkins.io-2026.key.
- systemd manages jenkins.service, so the server starts automatically on boot.

Conclusion

All the commands were executed successfully on Ubuntu 24.04 LTS and the evidence (command logs, screenshots and commit history) is committed in the GitHub repository github.com/theabhinav0231/devops-lab-1 under the docs/ directory.